

A guide to the arithmetic certificate for $K = \mathbf{Q}(\sqrt{-123482119})$, $p = 5$

Generated by `tools/certificate_guide.gp`

What this document is

This guide restates the content of `certificate.gp` in the notation of Section 2.5 of the paper. Each of the 18 entries certifies one value $D_x(e_j)$ of a secondary norm operator: it stores the unramified cyclic quintic L_x/K , the normalized generator σ_x , the pair (a', J) representing the torsion class e_j , the auxiliary divisor I' , the compactly represented element t , a residue prime for the sign check, and the expected norm class $[N_x(I')]$. The two defining equations are

$$(1 - \sigma_x)^2 I' + \text{div}(t) + i_x(J) = 0, \quad N_x(t) = a'.$$

Facts marked **Checked** are recomputed exactly by the generating script; the full verification is performed by the shared verifier `certificate/verify_certificate.c` as described in the README files of the `certificate` directory.

Notation and conventions

Ideals as matrices. An ideal of the ring of integers is stored by its *Hermite normal form* (HNF): an upper-triangular integer matrix whose columns are the coordinates, with respect to the fixed integral basis, of a $\mathbf{Z}$ -basis of the ideal. For example, the ideal $J = \begin{pmatrix} 2306 & 1001 \\ 0 & 1 \end{pmatrix}$ of Entry 1 below, over K with integral basis $(1, s)$, denotes the ideal $\mathbf{Z} 2306 + \mathbf{Z}(1001 + s)$. The determinant of the matrix is the norm of the ideal. Elements are likewise given by their coordinate vectors in the integral basis.

Characters. The header fixes three generators $\kappa_1, \kappa_2, \kappa_3$ of $\text{Cl}(K)$ (as HNF ideals). A character x on $\text{Cl}(K)/5$ is recorded by its value vector $(x(\bar{\kappa}_1), x(\bar{\kappa}_2), x(\bar{\kappa}_3))$; thus $(1, 0, 0)$ is the character dual to $\bar{\kappa}_1$. The column number j records which basis element e_j of $\text{Cl}(K)[5]$ the stored pair (a', J) represents.

Automorphisms. σ_x is a field automorphism of L_x . Writing θ for the chosen root of the stored absolute polynomial, so that $L_x = \mathbf{Q}(\theta)$, the automorphism is determined by the single value $\sigma_x(\theta)$, and the certificate stores the coordinate vector of $\sigma_x(\theta)$ in the integral basis of L_x .

Base field data

The base field is $K = \mathbf{Q}(s)$ with $s^2 - s + 30870530 = 0$, of discriminant -123482119 . Class group invariants $(270 \ 5 \ 5)$, class number 6750; the three stored generator ideals (HNF) are $\begin{pmatrix} 3701 & 407 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 5321 & 3325 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 5485 & 455 \\ 0 & 1 \end{pmatrix}$, and the torsion units are generated by -1 of order 2.

Entry 1: character a , column e_1

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + 3143y^3 + (13s + 203277)y^2 + (805s - 2808077)y + (3948s + 180371199)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 + 6290y^8 + 393995y^7 + 3449966y^6 + 1649817125y^5 \\ & + 28170766301y^4 + 638438154704y^3 + 104390489104063y^2 - 8166365 \\ & 85793047y + 33015651347624373 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ 0 \ 1 \ 0 \ -1 \ 0 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{101845979}{32603672280655888} + \left(-\frac{27721}{65207344561311776}\right)s, \quad J = \begin{pmatrix} 2306 & 1001 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2306$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $301379643833787 = 3^4 \cdot 3720736343627$:

$$\begin{pmatrix} 11162209030881 & 9694892882580 & 9429338425302 & 10539739703127 & 728631965825 & 4983667736004 & 484741371976 & 341684204281 & 7940809015466 & 7691241836001 \\ 0 & 3 & 0 & 0 & 0 & 1 & 0 & 0 & 2 & 1 \\ 0 & 0 & 3 & 0 & 0 & 0 & 1 & 2 & 1 & 2 \\ 0 & 0 & 0 & 3 & 2 & 2 & 2 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 2 factors with signed exponents of absolute value up to 1; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 4 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 2: character a , column e_2

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + 3143y^3 + (13s + 203277)y^2 + (805s - 2808077)y + (3948s + 180371199)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 + 6290y^8 + 393995y^7 + 3449966y^6 + 1649817125y^5 \\ & + 28170766301y^4 + 638438154704y^3 + 104390489104063y^2 - 8166365 \\ & 85793047y + 33015651347624373 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ 0 \ 1 \ 0 \ -1 \ 0 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{2039821869}{4265464134642730601} + \left(-\frac{58240}{4265464134642730601}\right)s, \quad J = \begin{pmatrix} 5321 & 3325 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5321$. **Checked:** $(a')^J = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $183134042912161 = 11 \cdot 12721 \cdot 1308745331$:

$$\begin{pmatrix} 183134042912161 & 20800793765058 & 138973395532151 & 1815266478758 & 117730674209745 & 130437536825493 & 124815487559744 & 50948764649726 & 114660011809634 & 149276735157783 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 2 factors with signed exponents of absolute value up to 2; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 3$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_2) = (0 \ 0 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 3: character a , column e_3

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + 3143y^3 + (13s + 203277)y^2 + (805s - 2808077)y + (3948s + 180371199)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 4y^9 + 6290y^8 + 393995y^7 + 3449966y^6 + 1649817125y^5 \\ + 28170766301y^4 + 638438154704y^3 + 104390489104063y^2 - 8166365 \\ 85793047y + 33015651347624373$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ 0 \ 1 \ 0 \ -1 \ 0 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{601351253}{4964587386703928125} + \left(-\frac{386132}{4964587386703928125}\right)s, \quad J = \begin{pmatrix} 5485 & 455 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5485$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $113043947459949 = 3^4 \cdot 11^3 \cdot 157 \cdot 331 \cdot 20177$:

$$\begin{pmatrix} 103805277741 & 79528556151 & 60665060555 & 14789234891 & 95688952133 & 19692499249 & 45724492741 & 75503706402 & 16594326331 & 57506767355 \\ 0 & 99 & 11 & 27 & 46 & 95 & 62 & 88 & 71 & 27 \\ 0 & 0 & 11 & 0 & 1 & 4 & 2 & 4 & 0 & 10 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 2 factors with signed exponents of absolute value up to 2; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_3) = (0 \ 4 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 4: character b , column e_1

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 3y^4 + (s - 809)y^3 + (5s + 42600)y^2 + (-1098s - 3684779)y + (-10350s - 58460997)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 6y^9 - 1608y^8 + 90056y^7 + 23897931y^6 + 144996679y^5 - \\ 58894974673y^4 - 1197446954596y^3 + 42622566601918y^2 + 11325779 \\ 63924682y + 6725221591477959$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ -1 \ 0 \ 1 \ -1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{101845979}{32603672280655888} + \left(-\frac{27721}{65207344561311776}\right) s, \quad J = \begin{pmatrix} 2306 & 1001 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2306$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $128889748995233 = 2557 \cdot 159499 \cdot 316031$:

$$\begin{pmatrix} 128889748995233 & 40302964010751 & 107693741560834 & 37161965696106 & 118244994313278 & 30446808581562 & 95073237225944 & 110267200967319 & 53139033007834 & 102926644032108 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 2 factors with signed exponents of absolute value up to 2; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 3$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 0 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 5: character b , column e_2

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 3y^4 + (s - 809)y^3 + (5s + 42600)y^2 + (-1098s - 3684779)y + (-10350s - 58460997)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 6y^9 - 1608y^8 + 90056y^7 + 23897931y^6 + 144996679y^5 - 58894974673y^4 - 1197446954596y^3 + 42622566601918y^2 + 1132577963924682y + 6725221591477959$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ -1 \ 0 \ 1 \ -1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{2039821869}{4265464134642730601} + \left(-\frac{58240}{4265464134642730601}\right) s, \quad J = \begin{pmatrix} 5321 & 3325 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5321$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $716420421540337 = 41 \cdot 3701 \cdot 4721337157$:

$$\begin{pmatrix} 716420421540337 & 439915674276760 & 486577769761015 & 378703171675192 & 54620729612907 & 426757902429438 & 568713151003147 & 3560740732264 & 559693418986395 & 62380647010835 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 2 factors with signed exponents of absolute value up to 1; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 3$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_2) = (0 \ 0 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 6: character b , column e_3

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 3y^4 + (s - 809)y^3 + (5s + 42600)y^2 + (-1098s - 3684779)y + (-10350s - 58460997)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 6y^9 - 1608y^8 + 90056y^7 + 23897931y^6 + 144996679y^5 - \\ & 58894974673y^4 - 1197446954596y^3 + 42622566601918y^2 + 11325779 \\ & 63924682y + 6725221591477959 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ -1 \ 0 \ 1 \ -1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{601351253}{4964587386703928125} + \left(-\frac{386132}{4964587386703928125}\right)s, \quad J = \begin{pmatrix} 5485 & 455 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5485$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $773848489785849 = 3^2 \cdot 131321 \cdot 654755641$:

$$\begin{pmatrix} 257949496595283 & 249684560401265 & 92865342864585 & 88881048392873 & 94241407971941 & 206258727450266 & 229796109771003 & 95221440722517 & 81115071193515 & 61039465987735 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 2 & 2 & 2 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 7 factors with signed exponents of absolute value up to 2; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_3) = (0 \ 0 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 7: character c , column e_1

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 2038y^3 + (-14s + 7)y^2 + 80212y + (28286s - 14143)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 4076y^8 + 4313868y^6 + 6377567943y^4 - 24443272561294y^2 + 24699442814327431$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \ -2 \ 1 \ 1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{101845979}{32603672280655888} + \left(-\frac{27721}{65207344561311776}\right)s, \quad J = \begin{pmatrix} 2306 & 1001 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2306$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1018586032545285 = 3^2 \cdot 5 \cdot 29^2 \cdot 43 \cdot 163 \cdot 3840017$:

$$\begin{pmatrix} 11707885431555 & 8897715785590 & 8731229078173 & 8967021363412 & 4428454790729 & 6458710182158 & 7347614000588 & 5456952104405 & 6953438502534 & 9992979416085 \\ 0 & 29 & 26 & 19 & 8 & 18 & 16 & 9 & 24 & 3 \\ 0 & 0 & 3 & 1 & 1 & 2 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 324 factors with signed exponents of absolute value up to 2852864096420; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_1) = (3 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 8: character c , column e_2

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 2038y^3 + (-14s + 7)y^2 + 80212y + (28286s - 14143)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 4076y^8 + 4313868y^6 + 6377567943y^4 - 24443272561294y^2 + 24699442814327431$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \ -2 \ 1 \ 1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{2039821869}{4265464134642730601} + \left(-\frac{58240}{4265464134642730601}\right)s, \quad J = \begin{pmatrix} 5321 & 3325 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5321$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $53073656002475 = 5^2 \cdot 43 \cdot 71 \cdot 695363983$:

$$\begin{pmatrix} 53073656002475 & 26349265848645 & 52254118441753 & 5162972146874 & 13650257953896 & 33232368743957 & 40154191976762 & 39969065660670 & 22965542274587 & 5470058977411 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 328 factors with signed exponents of absolute value up to 2125204679829; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (1 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 9: character c , column e_3

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 2038y^3 + (-14s + 7)y^2 + 80212y + (28286s - 14143)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 4076y^8 + 4313868y^6 + 6377567943y^4 - 24443272561294y^2 + 24699442814327431$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \ -2 \ 1 \ 1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{601351253}{4964587386703928125} + \left(-\frac{386132}{4964587386703928125}\right)s, \quad J = \begin{pmatrix} 5485 & 455 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5485$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $167516208537215 = 5 \cdot 187669 \cdot 178523047$:

$$\begin{pmatrix} 167516208537215 & 63543303819781 & 40891670051371 & 163132407498721 & 105857733906588 & 59150729847957 & 105987569427054 & 79278162390118 & 158050259564401 & 112876643350814 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 325 factors with signed exponents of absolute value up to 1528099554654; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \ 1 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 10: character $a + b$, column e_1

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + 2477y^3 + (s + 99677)y^2 + (-547s - 2176452)y + (25432s + 116411021)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 2y^9 + 4955y^8 + 194401y^7 + 1582723y^6 + 731003260y^5 - 1049871065y^4 + 109049721499y^3 + 38754726953157y^2 - 1365743682252975y + 33521132255139233$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ -1 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{101845979}{32603672280655888} + \left(-\frac{27721}{65207344561311776}\right)s, \quad J = \begin{pmatrix} 2306 & 1001 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2306$. **Checked:** $(a')^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $954790963349567 = 71^2 \cdot 189405071087$:

$$\begin{pmatrix} 13447760047177 & 7301202517667 & 5099431484620 & 1726471814904 & 9561197763960 & 7957130067557 & 7711065935835 & 638629510102 & 6546143098024 & 6232563125703 \\ 0 & 71 & 47 & 49 & 10 & 25 & 26 & 63 & 3 & 50 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 1080 factors with signed exponents of absolute value up to 25310787346386697130381438695021134989048594; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_1) = (1 \ 4 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 11: character $a + b$, column e_2

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + 2477y^3 + (s + 99677)y^2 + (-547s - 2176452)y + (25432s + 116411021)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 2y^9 + 4955y^8 + 194401y^7 + 1582723y^6 + 731003260y^5 - 1049871065y^4 + 109049721499y^3 + 38754726953157y^2 - 1365743682252975y + 33521132255139233$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ -1 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{2039821869}{4265464134642730601} + \left(-\frac{58240}{4265464134642730601}\right) s, \quad J = \begin{pmatrix} 5321 & 3325 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5321$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $705111402369537 = 3^2 \cdot 71 \cdot 424093 \cdot 2601931$:

$$\begin{pmatrix} 235037134123179 & 54781051842312 & 118746788783711 & 67082409614250 & 48814698257518 & 204769882102508 & 88737827282103 & 21600865656612 & 229950878499927 & 48241865615620 \\ 0 & 3 & 0 & 1 & 2 & 2 & 0 & 0 & 2 & 2 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 1085 factors with signed exponents of absolute value up to 12167129602129086946573171804864191419967684; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \ 0 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 12: character $a + b$, column e_3

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + 2477y^3 + (s + 99677)y^2 + (-547s - 2176452)y + (25432s + 116411021)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 2y^9 + 4955y^8 + 194401y^7 + 1582723y^6 + 731003260y^5 - 1049871065y^4 + 109049721499y^3 + 38754726953157y^2 - 1365743682252975y + 33521132255139233$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ -1 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{601351253}{4964587386703928125} + \left(-\frac{386132}{4964587386703928125}\right) s, \quad J = \begin{pmatrix} 5485 & 455 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5485$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $417818487826197 = 3^2 \cdot 46424276425133$:

$$\begin{pmatrix} 139272829275399 & 88713202440596 & 92755567741182 & 90177769625912 & 130026083504969 & 8390862915365 & 34713384729554 & 17208681903498 & 64312609986576 & 50418878540397 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 2 & 2 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 1085 factors with signed exponents of absolute value up to 11687433564430013207989635317620972294978905; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (1 \ 4 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 13: character $a + c$, column e_1

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 4223y^3 + (32s - 16)y^2 - 2600016y + (-2304s + 1152)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 8446y^8 + 12633697y^6 + 9651687328y^4 + 2208038365440y^2 + 163873614053376$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ -1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{101845979}{32603672280655888} + \left(-\frac{27721}{65207344561311776}\right)s, \quad J = \begin{pmatrix} 2306 & 1001 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2306$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $9464744484864 = 2^{13} \cdot 3^4 \cdot 257 \cdot 55501$:

$$\begin{pmatrix} 684660336 & 222154512 & 650284596 & 683220186 & 175996950 & 234377421 & 130277844 & 520380619 & 111932259 & 323728374 \\ 0 & 24 & 16 & 8 & 22 & 8 & 0 & 1 & 15 & 20 \\ 0 & 0 & 4 & 2 & 0 & 2 & 0 & 2 & 0 & 3 \\ 0 & 0 & 0 & 6 & 0 & 0 & 4 & 4 & 4 & 4 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 4 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 442 factors with signed exponents of absolute value up to 43099987836648487482; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (1 \ 1 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 14: character $a + c$, column e_2

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 4223y^3 + (32s - 16)y^2 - 2600016y + (-2304s + 1152)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 8446y^8 + 12633697y^6 + 9651687328y^4 + 2208038365440y^2 + 163873614053376$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ -1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{2039821869}{4265464134642730601} + \left(-\frac{58240}{4265464134642730601}\right)s, \quad J = \begin{pmatrix} 5321 & 3325 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5321$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $25775647135776 = 2^5 \cdot 3^2 \cdot 4621 \cdot 19367837$:

$$\begin{pmatrix} 1073985297324 & 5225662018 & 630948454836 & 1062010302135 & 354200203553 & 416677103064 & 535373255772 & 421369491208 & 285924777230 & 895865301630 \\ 0 & 2 & 0 & 0 & 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 12 & 11 & 8 & 7 & 7 & 0 & 9 & 9 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 442 factors with signed exponents of absolute value up to 40051445026885797606; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (3 \ 0 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 15: character $a + c$, column e_3

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 4223y^3 + (32s - 16)y^2 - 2600016y + (-2304s + 1152)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 8446y^8 + 12633697y^6 + 9651687328y^4 + 2208038365440y^2 + 163873614053376$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ -1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{601351253}{4964587386703928125} + \left(-\frac{386132}{4964587386703928125}\right)s, \quad J = \begin{pmatrix} 5485 & 455 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5485$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $91146205970432 = 2^{14} \cdot 5563122923$:

$$\begin{pmatrix} 356039867072 & 252209300936 & 66898099604 & 18048622616 & 229209357714 & 236761310544 & 2098126469 & 310097952939 & 319692912162 & 156875209885 \\ 0 & 8 & 4 & 0 & 3 & 4 & 4 & 1 & 0 & 0 \\ 0 & 0 & 4 & 0 & 3 & 3 & 0 & 3 & 3 & 0 \\ 0 & 0 & 0 & 8 & 0 & 0 & 6 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 437 factors with signed exponents of absolute value up to 30658864764070769427; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (4 \ 2 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 16: character $b + c$, column e_1

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 1594y^3 + (32s - 16)y^2 - 2525895y + (-2592s + 1296)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 3188y^8 - 2510954y^6 + 39663975724y^4 + 1259095111857y^2 + 207402542786304$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-2 \ 2 \ -1 \ 1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{101845979}{32603672280655888} + \left(-\frac{27721}{65207344561311776}\right)s, \quad J = \begin{pmatrix} 2306 & 1001 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2306$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $80312990617728 = 2^7 \cdot 3^4 \cdot 11 \cdot 101 \cdot 6972311$:

$$\begin{pmatrix} 557729101512 & 349832153790 & 462355049748 & 274706643287 & 342513048316 & 365118894658 & 111611279999 & 355101108901 & 50239291891 & 23562060756 \\ 0 & 18 & 0 & 1 & 2 & 0 & 8 & 4 & 10 & 9 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 254 factors with signed exponents of absolute value up to 4061008623; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (3 \ 3 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 17: character $b + c$, column e_2

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 1594y^3 + (32s - 16)y^2 - 2525895y + (-2592s + 1296)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 3188y^8 - 2510954y^6 + 39663975724y^4 + 1259095111857y^2 + 207402542786304$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-2 \ 2 \ -1 \ 1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{2039821869}{4265464134642730601} + \left(-\frac{58240}{4265464134642730601}\right)s, \quad J = \begin{pmatrix} 5321 & 3325 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5321$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $461557383369984 = 2^8 \cdot 3^{10} \cdot 11^2 \cdot 252341$:

$$\begin{pmatrix} 1798686648 & 666047448 & 607042272 & 655620738 & 108376403 & 37304866 & 299819624 & 970450028 & 1288038357 & 920414049 \\ 0 & 1782 & 1716 & 501 & 709 & 859 & 488 & 369 & 599 & 294 \\ 0 & 0 & 12 & 6 & 8 & 9 & 2 & 9 & 4 & 6 \\ 0 & 0 & 0 & 6 & 3 & 1 & 4 & 3 & 5 & 2 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 259 factors with signed exponents of absolute value up to 2003087558; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (1 \ 1 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 18: character $b + c$, column e_3

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 1594y^3 + (32s - 16)y^2 - 2525895y + (-2592s + 1296)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 3188y^8 - 2510954y^6 + 39663975724y^4 + 1259095111857y^2 + 207402542786304$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-2 \ 2 \ -1 \ 1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{601351253}{4964587386703928125} + \left(-\frac{386132}{4964587386703928125}\right)s, \quad J = \begin{pmatrix} 5485 & 455 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5485$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $457970075887104 = 2^9 \cdot 3^2 \cdot 11^3 \cdot 74670073$:

$$\begin{pmatrix} 19712899272 & 7067514300 & 19703872672 & 12268240548 & 5325940829 & 78057772 & 2095614532 & 17306151784 & 6427559407 & 5934028045 \\ 0 & 132 & 44 & 76 & 66 & 44 & 56 & 33 & 120 & 126 \\ 0 & 0 & 22 & 0 & 0 & 8 & 0 & 21 & 12 & 6 \\ 0 & 0 & 0 & 2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 258 factors with signed exponents of absolute value up to 3995280885; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (2 \ 3 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.